

# Supplementary Note

## Auxiliary Enclosures, Multiscale Expansions, and Numerical Diagnostics

*for Order-Dependent Triangular Limits and  
Double-Scaling Transitions for the Logarithmic Integral*

Independent research draft

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This note collects mathematically valid but non-load-bearing developments separated from the main article to preserve its focus on the order-dependent limit and its fixed-, moving-, and vanishing-window scaling transitions. The material includes elementary enclosures for the finite-window limit, an exact incomplete-gamma tail, local and multiscale expansions with signed remainders, parameter tuning, extended numerical diagnostics, a classical renormalized zero-window identity, and reproducibility formulas. These results support interpretation and independent verification but are not used to establish the central double-scaling theorems.

## Abstract

This supplementary note records auxiliary results for the logarithmic-integral triangular family studied in the accompanying main article. First, the limiting window difference  $D(x) = \text{Li}(x) - \text{Li}(x/e)$  receives nested elementary enclosures and an exact positive incomplete-gamma remainder. Second, local logarithmic expansions are assembled into a multiscale representation of the offset logarithmic integral with a signed remainder, certified odd/even bracketing, and a global geometric error bound; its sharp full-step threshold is  $a = 1/2$ . Further sections give a transparent work model, numerical diagnostics, the classical renormalized zero-window limit, and stable recurrences used to reproduce the computations. No separate novelty claim is made for the classical Ei identity or its power series.

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## S1 Purpose and finite-window notation

The main article proves that the prescribed triangular sequence converges, for  $x > e$ , to

$$D(x) := \text{Li}(x) - \text{Li}(x/e) = x \sum_{m \geq 0} \frac{\gamma(m+1, 1)}{L^{m+1}}, \quad L := \log x.$$

The results below are consequences, alternative decompositions, diagnostics, or implementation details rather than inputs to that theorem. Cross-references to the coupled and scaling theorems refer to the accompanying main article.

### S1.1 Elementary finite-window enclosures

**Corollary S1.1** (Elementary two-sided enclosures). *Let  $x > e$ , put  $L = \log x$ , and define*

$$a_m := \gamma(m+1, 1) = m! \left( 1 - \frac{1}{e} \sum_{j=0}^m \frac{1}{j!} \right), \quad b_m := \frac{e-1}{e(m+1)}.$$

For an integer  $K \geq 0$ , set

$$P_K(x) := x \sum_{m=0}^K \frac{a_m}{L^{m+1}}, \tag{S1}$$

$$U_K(x) := \frac{e-1}{e} x \log \frac{L}{L-1} + x \sum_{m=1}^K \frac{a_m - b_m}{L^{m+1}}, \tag{S2}$$

where the second sum is empty when  $K = 0$ . Then

$$\boxed{P_K(x) < P_{K+1}(x) < D(x) < U_{K+1}(x) < U_K(x), \quad x > e.} \tag{S3}$$

The enclosure width is the elementary expression

$$\boxed{U_K(x) - P_K(x) = \frac{e-1}{e} x \left[ \log \frac{L}{L-1} - \sum_{m=0}^K \frac{1}{(m+1)L^{m+1}} \right].} \tag{S4}$$

For fixed  $K$ , the lower and upper errors satisfy

$$D(x) - P_K(x) \sim \frac{x a_{K+1}}{L^{K+2}}, \quad U_K(x) - D(x) \sim \frac{x(b_{K+1} - a_{K+1})}{L^{K+2}} \quad (x \rightarrow \infty).$$

Thus every additional correction improves the relative order by one power of  $1/\log x$ .

*Proof.* Theorem 4.3 gives

$$D(x) = x \sum_{m=0}^{\infty} \frac{a_m}{L^{m+1}},$$

so positivity immediately yields  $P_K(x) < D(x)$ . For  $m \geq 1$ , the functions  $t^m$  and  $e^{-t}$  are strictly oppositely monotone on  $[0, 1]$ . The elementary reversed Chebyshev inequality gives

$$a_m = \int_0^1 t^m e^{-t} dt < \left( \int_0^1 t^m dt \right) \left( \int_0^1 e^{-t} dt \right) = \frac{e-1}{e(m+1)} = b_m.$$

For completeness, this inequality follows by integrating  $(s^m - t^m)(e^{-s} - e^{-t}) < 0$  over  $[0, 1]^2$ . Since  $a_0 = b_0$  and

$$\frac{e-1}{e} \log \frac{L}{L-1} = \sum_{m=0}^{\infty} \frac{b_m}{L^{m+1}},$$

formula (S2) may be rewritten as

$$U_K(x) = x \sum_{m=0}^K \frac{a_m}{L^{m+1}} + x \sum_{m=K+1}^{\infty} \frac{b_m}{L^{m+1}}.$$

Consequently,

$$U_K(x) - D(x) = x \sum_{m=K+1}^{\infty} \frac{b_m - a_m}{L^{m+1}} > 0.$$

Moreover,

$$P_{K+1} - P_K = \frac{x a_{K+1}}{L^{K+2}} > 0, \quad U_{K+1} - U_K = \frac{x(a_{K+1} - b_{K+1})}{L^{K+2}} < 0,$$

which proves the nesting. Subtracting (S1) from the rewritten form of  $U_K$  and summing the logarithmic tail gives (S4). The two asymptotic statements follow from the first omitted terms of the corresponding convergent series.  $\square$

The first enclosure is

$$\frac{e-1}{e} \frac{x}{L} < D(x) < \frac{e-1}{e} x \log \frac{L}{L-1}, \quad (\text{S5})$$

and correcting one coefficient gives the sharper elementary pair

$$x \left( \frac{e-1}{eL} + \frac{e-2}{eL^2} \right) < D(x) < \frac{e-1}{e} x \log \frac{L}{L-1} + \frac{e-3}{2e} \frac{x}{L^2}. \quad (\text{S6})$$

These bounds are certified for every  $x > e$ ; they are not merely large- $x$  comparisons. The logarithmic upper bound is the resummation of a deliberately coarser moment majorant, while the successive corrections replace its first coefficients by the exact incomplete-gamma moments.

**Remark S1.2** (Meaning of “absolutely”). For each  $N$ ,  $T_N(x; N)$  is finite. Absolute convergence in Theorem 4.3 refers to the limiting incomplete-gamma series, while convergence of  $T_N(x; N)$  refers to a sequence of different finite triangular sums. It does not repair the fixed- $a$  series of Proposition 3.1.

## S2 The incomplete-gamma bridge and exact tail

The limit in Theorem 4.3 is itself a particularly simple anchored expansion. Let

$$S_M(x) := x \sum_{m=0}^M \frac{\gamma(m+1, 1)}{L^{m+1}}.$$

**Proposition S2.1** (Exact positive remainder). *For  $x > e$ ,*

$$D(x) - S_M(x) = \frac{x}{L^{M+1}} \int_0^1 \frac{t^{M+1} e^{-t}}{L-t} dt > 0, \quad (\text{S7})$$

and

$$0 < D(x) - S_M(x) \leq L^{-(M+1)} D(x). \quad (\text{S8})$$

Thus  $S_M(x)$  increases monotonically to  $D(x)$ .

*Proof.* Use the finite geometric identity

$$\frac{1}{L-t} = \sum_{m=0}^M \frac{t^m}{L^{m+1}} + \frac{t^{M+1}}{L^{M+1}(L-t)}$$

inside (11). This gives (S7). The remainder integrand is the integrand defining  $D(x)$  multiplied by  $(t/L)^{M+1} \leq L^{-(M+1)}$ , proving (S8).  $\square$

The same argument works on any finite logarithmic interval. For  $x > e$  it gives the anchored identity

$$\text{Li}(x) = \text{Li}(e) + x \sum_{m=0}^{\infty} \frac{\gamma(m+1, L-1)}{L^{m+1}}, \quad (\text{S9})$$

where

$$\gamma(m+1, L-1) = \int_0^{L-1} t^m e^{-t} dt.$$

The ratio controlling the tail is  $(L-1)/L$ . Formula (S9) is convergent for each fixed  $x > e$ , but this ratio approaches one as  $x \rightarrow \infty$ . The multiscale construction below replaces the single long interval by shorter pieces whose ratio is bounded independently of  $x$ .

### S3 A local alternating expansion and its sharp threshold

Let  $0 < a < 1$  and  $x > 1$ . Directly from the definition,

$$\text{Li}(x) - \text{Li}(x^a) = \int_{x^a}^x \frac{du}{\log u}.$$

Set  $u = x^a s$ . Then

$$\text{Li}(x) - \text{Li}(x^a) = x^a \int_1^{x^{1-a}} \frac{ds}{aL + \log s}. \quad (\text{S10})$$

The formal geometric expansion of the denominator gives

$$\text{Li}(x) - \text{Li}(x^a) = x^a \sum_{k=0}^{\infty} \frac{(-1)^k}{(aL)^{k+1}} \int_1^{x^{1-a}} (\log s)^k ds. \quad (\text{S11})$$

**Proposition S3.1** (Exact local remainder). *For every  $N \geq 0$ , define the truncation of the right side of (S11) by  $Q_N(x; a)$ . Then*

$$\begin{aligned} \text{Li}(x) - \text{Li}(x^a) - Q_N(x; a) \\ = (-1)^{N+1} \frac{x^a}{(aL)^{N+1}} \int_1^{x^{1-a}} \frac{(\log s)^{N+1}}{aL + \log s} ds. \end{aligned} \quad (\text{S12})$$

If  $a > 1/2$  and  $q = (1-a)/a$ , then

$$|\text{Li}(x) - \text{Li}(x^a) - Q_N(x; a)| \leq q^{N+1} (\text{Li}(x) - \text{Li}(x^a)). \quad (\text{S13})$$

*Proof.* Apply

$$\frac{1}{1+z} = \sum_{k=0}^N (-1)^k z^k + (-1)^{N+1} \frac{z^{N+1}}{1+z}$$

with  $z = (\log s)/(aL)$  inside (S10). This gives (S12). On the integration interval,

$$0 \leq z \leq \frac{(1-a)L}{aL} = q.$$

If  $q < 1$ , the absolute remainder integrand is the exact integrand multiplied by  $z^{N+1} \leq q^{N+1}$ , proving (S13).  $\square$

**Proposition S3.2** (Threshold at one half). *For a full scale step  $x^a \rightarrow x$ , the series (S11) is absolutely convergent when  $a > 1/2$ , conditionally convergent when  $a = 1/2$ , and divergent when  $0 < a < 1/2$ .*

*Proof.* Put  $b = (1 - a)L$  and substitute  $v = \log s$  in the moment:

$$\int_1^{x^{1-a}} (\log s)^k ds = \int_0^b v^k e^v dv \sim \frac{e^b b^{k+1}}{k+1}. \quad (\text{S14})$$

The asymptotic follows from the standard endpoint-moment lemma: for continuous  $f$  at  $b > 0$ ,  $\int_0^b v^k f(v) dv \sim f(b) b^{k+1}/(k+1)$ . Hence the magnitude of the  $k$ th term of (S11) is asymptotic to a positive constant times

$$\frac{1}{k+1} \left( \frac{1-a}{a} \right)^{k+1}.$$

If  $a > 1/2$ , the ratio is below one and absolute convergence follows. If  $a < 1/2$ , the terms do not tend to zero. At  $a = 1/2$  their magnitudes are asymptotic to a constant divided by  $k+1$ ; absolute convergence fails. Conditional convergence follows either from the alternating-series test after eventual monotonicity, or directly by integrating the pointwise geometric series on every truncated interval and using the exact remainder (S12), which tends to zero at the harmonic endpoint rate.  $\square$

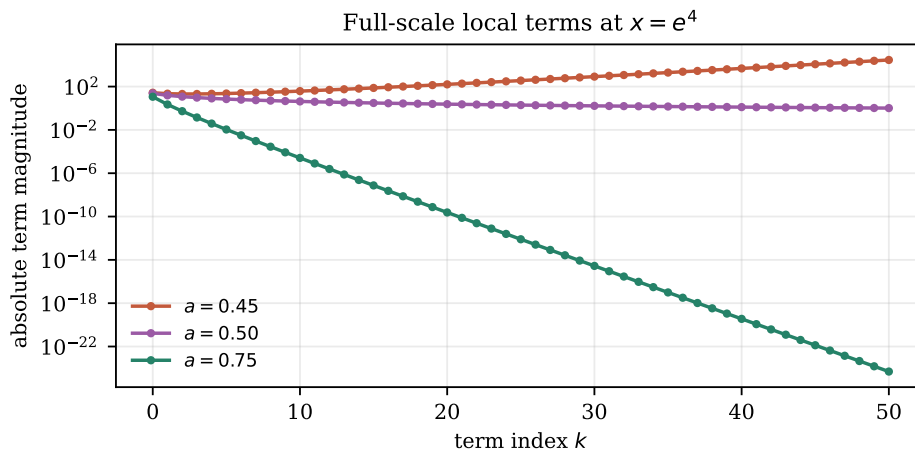


Figure S1: Absolute local-series terms for a full scale step at  $x = e^4$ . For  $a = 0.75$  the terms decay geometrically; at  $a = 0.50$  they decay only harmonically; and for  $a = 0.45$  they eventually grow. These are the three regimes of Proposition S3.2.

The truncation-level view in Figure S1 makes the boundary  $a = 1/2$  quantitative. Above it, each extra term buys an essentially fixed factor of accuracy. At the boundary, improvement persists but becomes slow and conditional. Below it, the term test fails, so increasing the truncation order cannot produce a convergent full-step expansion.

## S4 The multiscale expansion

Let  $x > 2$ ,  $U = \log x$ , and  $b = \log 2$ . Fix  $a \in (1/2, 1)$  and define

$$U_j := \max\{b, a^j U\}, \quad J := \left\lceil \frac{\log(b/U)}{\log a} \right\rceil. \quad (\text{S15})$$

Then  $U_0 = U$ ,  $U_J = b$ , and  $U_j > b$  for  $j < J$ . Put  $\Delta_j := U_j - U_{j+1}$ . The logarithmic integral telescopes as

$$\text{Li}(x) = \sum_{j=0}^{J-1} \int_{U_{j+1}}^{U_j} \frac{e^s}{s} ds. \quad (\text{S16})$$

Every full step has  $U_{j+1} = aU_j$ ; the final clipped step is shorter. Consequently

$$0 \leq \frac{\Delta_j}{U_{j+1}} \leq q := \frac{1-a}{a} < 1. \quad (\text{S17})$$

For  $N \geq 0$ , define

$$M_N(x; a) := \sum_{j=0}^{J-1} e^{U_{j+1}} \sum_{k=0}^N \frac{(-1)^k}{U_{j+1}^{k+1}} \int_0^{\Delta_j} t^k e^t dt. \quad (\text{S18})$$

**Theorem S4.1** (Multiscale expansion and global remainder). *For every  $x > 2$  and  $1/2 < a < 1$ ,*

$$\boxed{\text{Li}(x) = \sum_{j=0}^{J-1} e^{U_{j+1}} \sum_{k=0}^{\infty} \frac{(-1)^k}{U_{j+1}^{k+1}} \int_0^{\Delta_j} t^k e^t dt.} \quad (\text{S19})$$

*Each inner series is absolutely convergent. The exact truncation error is*

$$\text{Li}(x) - M_N(x; a) = (-1)^{N+1} \sum_{j=0}^{J-1} \frac{e^{U_{j+1}}}{U_{j+1}^{N+1}} \int_0^{\Delta_j} \frac{t^{N+1} e^t}{U_{j+1} + t} dt. \quad (\text{S20})$$

*In particular all segment remainders have the same sign, and*

$$\boxed{|\text{Li}(x) - M_N(x; a)| \leq q^{N+1} \text{Li}(x), \quad q = \frac{1-a}{a}.} \quad (\text{S21})$$

*Thus even  $N$  gives an overestimate and odd  $N$  an underestimate.*

*Proof.* In the  $j$ th integral of (S16), set  $s = U_{j+1} + t$ :

$$\int_{U_{j+1}}^{U_j} \frac{e^s}{s} ds = e^{U_{j+1}} \int_0^{\Delta_j} \frac{e^t}{U_{j+1} + t} dt. \quad (\text{S22})$$

Use the finite identity

$$\begin{aligned} \frac{1}{U_{j+1} + t} &= \sum_{k=0}^N \frac{(-1)^k t^k}{U_{j+1}^{k+1}} \\ &\quad + (-1)^{N+1} \frac{t^{N+1}}{U_{j+1}^{N+1} (U_{j+1} + t)}. \end{aligned}$$

After multiplying by  $e^{U_{j+1}+t}$  and integrating, the finite sum is the  $j$ th portion of  $M_N$  and the last term is the  $j$ th portion of (S20). Summing over the finite set of scales proves the exact remainder formula.

For absolute convergence, the absolute  $k$ th moment contribution is bounded by the segment integral in (S22) times  $(\Delta_j/U_{j+1})^k$ ; alternatively one may apply uniform convergence of the geometric series on  $0 \leq t/U_{j+1} \leq q < 1$ . Finally, the absolute remainder integrand in each segment is the exact positive integrand multiplied by  $(t/U_{j+1})^{N+1} \leq q^{N+1}$ . Summing and using (S16) proves (S21). The sign statement follows directly from (S20).  $\square$

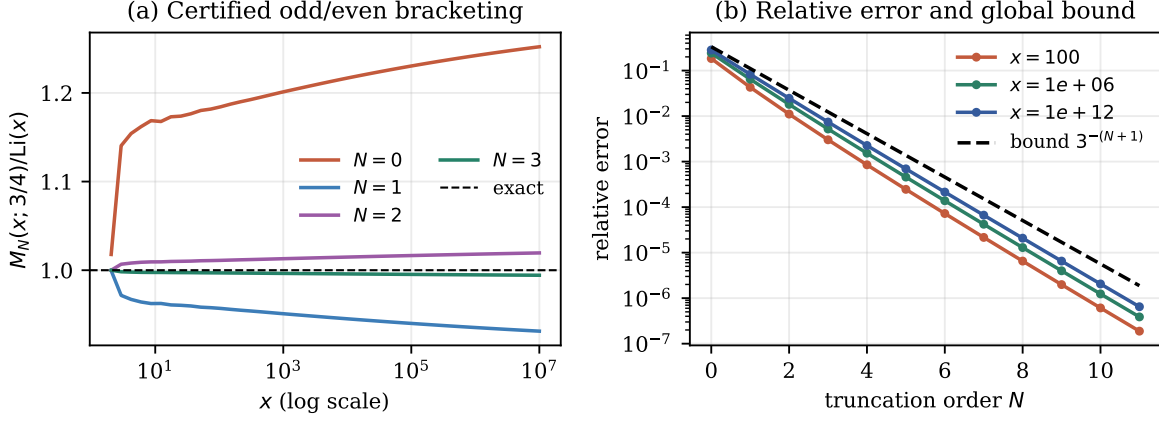


Figure S2: Certified multiscale truncations with  $a = 3/4$ . Left: even orders lie above and odd orders below  $\text{Li}(x)$  over a wide  $x$ -range. Right: the measured relative errors for three values of  $x$  remain beneath the uniform envelope  $3^{-(N+1)}$  from Theorem S4.1.

**Corollary S4.2** (Certified bracketing). *For every  $r \geq 0$ ,*

$$M_{2r+1}(x; a) \leq \text{Li}(x) \leq M_{2r}(x; a),$$

*and each endpoint is within the relative error prescribed by (S21).*

Figure S2 shows the two features that distinguish the multiscale construction from a merely convergent numerical series. Parity supplies a rigorous enclosure, and the same geometric relative-error guarantee holds independently of the size of  $x$ . The lowest orders visibly form a wide bracket; successive truncations tighten it at the certified factor  $q = 1/3$  or faster.

**Remark S4.3** (The endpoint  $x = 2$ ). At  $x = 2$ ,  $J = 0$ , the scale sum is empty, and both sides are zero. Thus the theorem extends to  $x = 2$  by the natural empty-sum convention, although the relative interpretation is then vacuous.

## S5 Recursion, boundary behavior, and parameter tuning

The first scale of Theorem S4.1 may be written in the original variable as

$$\text{Li}(x) = \text{Li}(x^a) + x^a \sum_{k=0}^{\infty} \frac{(-1)^k}{(a \log x)^{k+1}} \int_1^{x^{1-a}} (\log t)^k dt, \quad (\text{S23})$$

provided  $a > 1/2$ . Iterating (S23) until the lower scale reaches 2 gives the multiscale formula, with a clipped last interval. Therefore the recursive and multiscale forms are not competing approximations: they are the same telescoping construction presented at different levels of expansion.

At  $a = 1/2$ , any full geometric scale has endpoint ratio one. Its local moment series is conditionally convergent by Proposition S3.2; because the number of scales is finite, the corresponding multiscale expression is conditionally convergent whenever full steps occur. If  $0 < a < 1/2$  and  $x$  is large enough that at least one full step occurs, that segment's terms fail to tend to zero, so the multiscale moment expansion diverges. A clipped interval can be shorter and may converge even below  $1/2$ ; thus the sharp statement concerns each full geometric step, not every possible short terminal interval.

### S5.1 A transparent work model

For large  $U/b$ , the number of scales is approximately

$$J \approx \frac{\log(U/b)}{-\log a}. \quad (\text{S24})$$

If each scale uses  $N + 1$  moments, a simple operation proxy is  $W = J(N + 1)$ . The error bound is  $q^{N+1}$ , so the logarithmic error reduction per unit work is, apart from the  $x$ -dependent factor  $\log(U/b)$ , proportional to

$$F(a) := (-\log a) \left[ -\log \left( \frac{1-a}{a} \right) \right]. \quad (\text{S25})$$

The stationary equation  $F'(a) = 0$  is

$$(1-a) \log(1-a) = (2-a) \log a. \quad (\text{S26})$$

Equivalently,  $(1-a) \log((1-a)/a^2) = a \log a$ . Numerically, the maximizer is approximately  $a = 0.758515$ . At  $a = 3/4$ , the value of  $F$  is within approximately 0.094% of its maximum. Thus  $a = 3/4$  is effectively optimal for this particular equal-cost moment model and also has the convenient contraction  $q = 1/3$ .

This does not make  $3/4$  universally optimal. The cost of evaluating moments, reuse between scales, floating-point cancellation, and a target absolute rather than relative error all change the objective. Equation (S25) is therefore a tuning guide, not a universal theorem about implementation.

## S6 Numerical diagnostics

The following computations use direct high-precision evaluation of the finite expressions and of the defining integrals. They are diagnostics, not substitutes for the proofs.

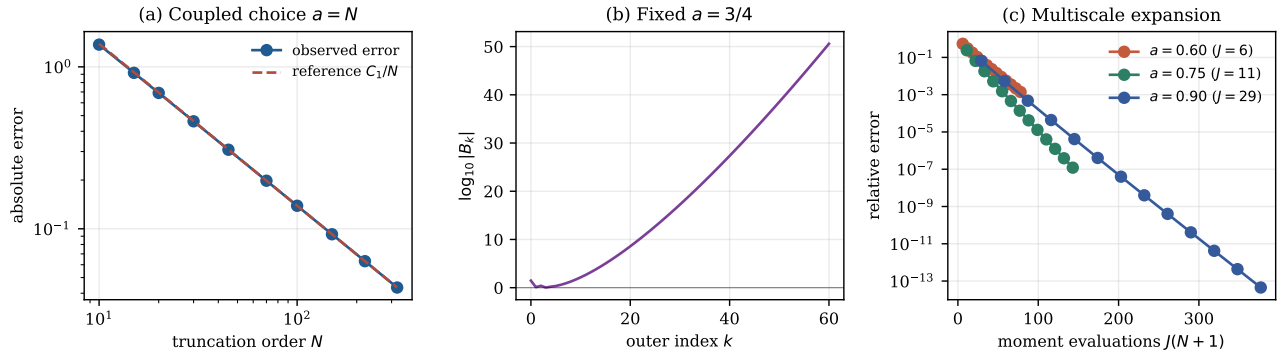


Figure S3: Three diagnostic views. Left: the diagonal error at  $x = 100$  approaches  $C_1(100)/N$  from Theorem 5.6 and stays below the explicit bound of Theorem 5.2. Center: fixed  $a = 3/4$  outer blocks eventually grow factorially. Right: multiscale relative error versus the total moment count  $J(N + 1)$  at  $x = 10^6$ ;  $a = 3/4$  balances depth and contraction well in this cost model.

Table S1 illustrates both coefficient limits in Theorem 5.6: the third column converges to  $C_1(x)$ , while the fourth converges to  $C_2(x)$  and visibly leaves an  $O(N^{-1})$  scaled residual, corresponding to the theorem's  $O_x(N^{-3})$  remainder.

Table S1: Scaled first- and second-order diagonal residuals for  $T_N(x; N)$ , with  $D(x) = \text{Li}(x) - \text{Li}(x/e)$ .

| $x$    | $N$ | $N[T_N - D]$ | $N^2[T_N - D - C_1/N]$ |
|--------|-----|--------------|------------------------|
| 10     | 50  | 3.14935347   | -0.1450134             |
|        | 100 | 3.15086165   | -0.1392088             |
|        | 200 | 3.15157333   | -0.1360806             |
|        | 400 | 3.15191760   | -0.1344523             |
| 100    | 50  | 13.85772184  | -1.669288              |
|        | 100 | 13.87446869  | -1.663891              |
|        | 200 | 13.88280155  | -1.661209              |
|        | 400 | 13.88695792  | -1.659872              |
| $10^6$ | 50  | 41798.4684   | -7013.182              |
|        | 100 | 41868.8420   | -6989.003              |
|        | 200 | 41903.8471   | -6976.991              |
|        | 400 | 41921.3045   | -6971.003              |

Theorem 5.6 predicts the limiting values  $C_1(10) = 3.15225373395$ ,  $C_1(100) = 13.8911075951$ , and  $C_1(10^6) = 41938.7320511$  for the third column. The displayed finite- $N$  values approach these constants with the sign and first-order rate asserted by the theorem. The fourth column approaches the second-order constants  $C_2(10) \approx -0.1327777862$ ,  $C_2(100) \approx -1.658537447$ , and  $C_2(10^6) \approx -6965.028591$ .

For the multiscale test at  $x = 10^6$ , the scale counts are  $J = 6, 11, 29$  for  $a = 0.6, 0.75, 0.9$ , respectively. At  $N = 11$ , the observed relative errors are approximately  $2.21 \times 10^{-3}$ ,  $3.87 \times 10^{-7}$ , and  $4.34 \times 10^{-13}$ . The last choice contracts most strongly per moment but requires nearly five times as many scales as  $a = 0.6$ . Plotting against total moments, rather than  $N$  alone, makes the tradeoff visible.

### S6.1 Fixed- and vanishing-window scaling diagnostics

The following tables and figures collect the extended checks for the Borel model, the shifted model, and the first vanishing-window correction. They were moved from the main article because they verify rather than establish the scaling theorems.

Table S2: Anchored vanishing- $\lambda$  scaling for  $\delta = \log 2$ . The entries are the left side of (54), evaluated from the exact recurrence recorded in the supplementary note;  $N$  is the nearest integer to  $\mu^2/(u\delta^2)$ .

| $u$  | $\mu = 16$  | $\mu = 32$  | $\mu = 64$  | $\Psi_{\log 2}(u)$ |
|------|-------------|-------------|-------------|--------------------|
| 0.25 | -0.01435975 | -0.01952282 | -0.02210358 | -0.02468409        |
| 1    | -0.00532576 | -0.02405958 | -0.03342199 | -0.04277376        |
| 4    | 0.13470481  | 0.07377234  | 0.04359108  | 0.01336457         |

The first correction in Proposition 5.25 is also visible directly. For  $\delta = \log 2$ , the predicted constants and the scaled residuals  $\mu(F_\mu - \Psi_{\log 2})$  are

| $u$  | $K_{\log 2}(u)$ | $\mu = 16$ | $\mu = 32$ | $\mu = 64$ |
|------|-----------------|------------|------------|------------|
| 0.25 | 0.16514058      | 0.16519    | 0.16516    | 0.16515    |
| 1    | 0.59821452      | 0.59917    | 0.59885    | 0.59851    |
| 4    | 1.9320865       | 1.94144    | 1.93305    | 1.93450    |

The agreement confirms both the  $O(\mu^{-1})$  rate and the positive sign of the correction.

Figure S5 illustrates the two proved limiting statements in a common set of coordinates. The separation between the colored curves and the black curve is a finite- $\delta$  effect; it decreases as the threshold is

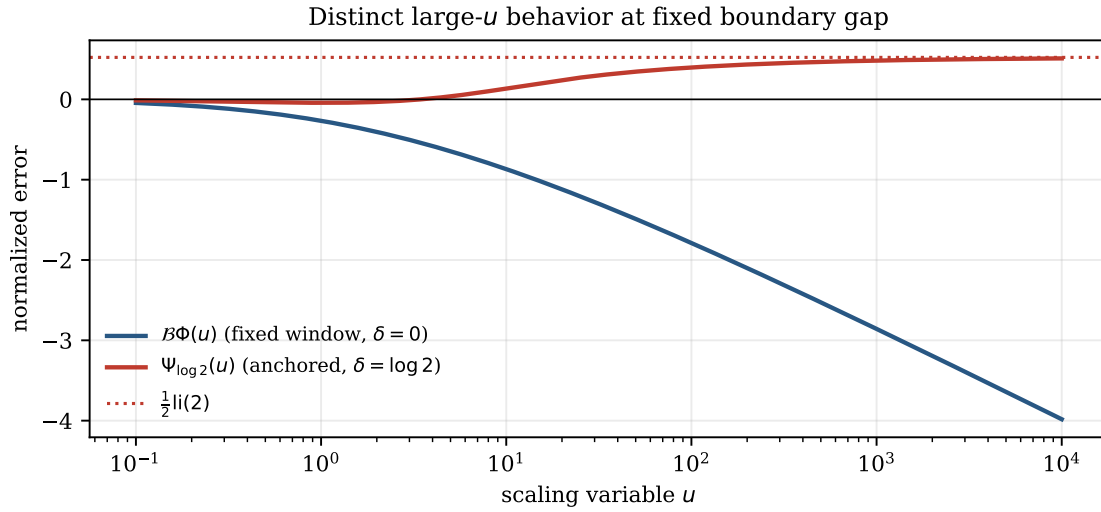


Figure S4: The large- $u$  phase distinction in Corollary 5.23. For the anchored gap  $\delta = \log 2$ , the shifted model crosses zero and approaches  $\frac{1}{2} \text{li}(2)$ , while the endpoint model  $\mathcal{B}\Phi(u)$  decreases like  $-\frac{1}{2} \log u$ . Hence  $\Psi_\delta \rightarrow \mathcal{B}\Phi$  as  $\delta \downarrow 0$  is pointwise in  $u$  but not uniform on the positive half-line.

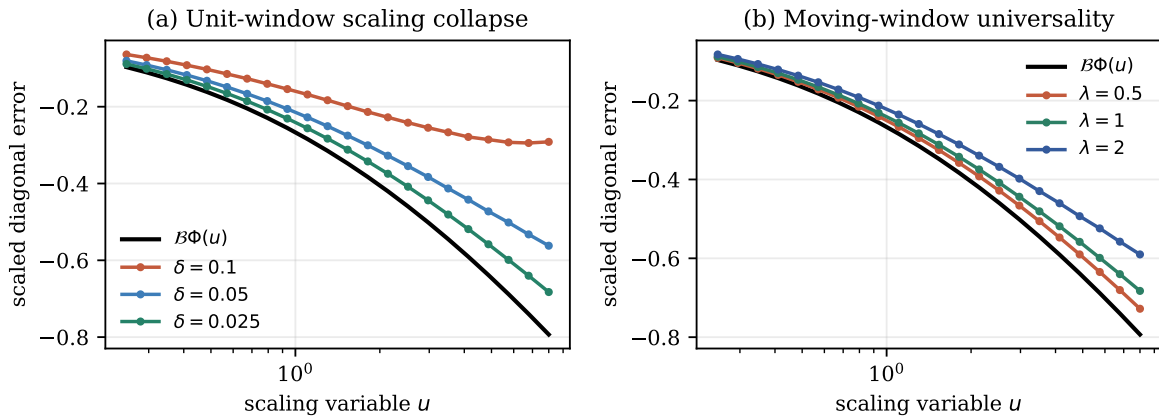


Figure S5: The double-scaling limit across truncation levels and window lengths. Left: for the unit window, decreasing  $\delta = L - 1$  drives the finite- $N$  curves toward the exact Borel model. Right: at  $\delta = 0.025$ , the couplings  $\lambda = 1/2, 1, 2$  approach the same limiting curve after the normalization in Theorem 5.18.

approached. The right panel is the numerical signature of the  $\lambda$ -independence proved in Theorem 5.18. The figure is not used to assert the unproved global  $O_{u,\lambda}(\delta)$  rate discussed in Remark 5.17.

Table S3 illustrates the approach to the limit proved in Theorem 5.16. Here  $F(N, \delta) = e[T_N(x; N) - D(x)]/x$  and  $N$  is the nearest integer to  $1/(u\delta^2)$ . The residual approximately halves with  $\delta$ , including at  $u = 4$  and  $8$ , well beyond the small- $u$  linear regime. This observed behavior motivates, but does not prove, the  $O_u(\delta)$  rate discussed in Remark 5.17.

Table S3: Numerical approach to the double-scaling limit in Theorem 5.16.

| $u$  | $\mathcal{B}\Phi(u)$ | $F(0.1)$   | $F(0.05)$ | $F(0.025)$ | $F(0.0125)$ |
|------|----------------------|------------|-----------|------------|-------------|
| 0.25 | -0.096876            | -0.063960  | -0.080510 | -0.088717  | —           |
| 0.5  | -0.165482            | -0.105169  | -0.135644 | -0.150646  | —           |
| 1    | -0.266727            | -0.159956  | -0.214346 | -0.240800  | -0.253831   |
| 4    | -0.581356            | -0.276345  | -0.436541 | -0.510979  | -0.546689   |
| 8    | -0.793236            | -0.291564* | -0.561991 | -0.682651  | -0.739237   |

The dashes indicate entries not computed for the displayed table; they do not mark a mathematical exception or a change in the limiting behavior.

At  $u = 1$ , first-order Richardson extrapolation gives  $-0.267254$  and  $-0.266861$ , compared with  $\mathcal{B}\Phi(1) = -0.266727$ ; the residual ratio is approximately 0.256, consistent with a smooth expansion in  $\delta$ . The observed leading  $O(\delta)$  coefficient is not the  $j = 1$  contribution alone: higher pole coefficients contribute at the same order, as required for a genuinely resummed limit.

The starred  $u = 8$ ,  $\delta = 0.1$  entry uses the tie-break  $N = 12$ . Its realized parameter is  $u_{\text{eff}} = 1/(12\delta^2) = 25/3$ , for which  $\mathcal{B}\Phi(u_{\text{eff}}) = -0.806733$ ; residual comparisons for that cell use this effective target. Every other displayed cell has integral  $1/(u\delta^2)$ .

The  $u = 8$  row is already nonperturbative: the first three formal terms are  $-4, +48, -1280$ , so optimal truncation occurs at the first term. The row therefore illustrates the proved limit in a regime where ordinary low-order truncation of the formal series is already ineffective. Additional large- $u$  illustrations give

$$\mathcal{B}\Phi(100) = -1.787936, \quad \mathcal{B}\Phi(1000) = -2.857836.$$

At  $u = 100$ , using  $\delta = 0.02, 0.01, 0.005, 0.0025$ , the residual ratios are 0.4612, 0.4788, 0.4889 after the first point; at  $u = 1000$ , using  $\delta = 0.01, 0.005, 0.0025, 0.00125$ , they are 0.4384, 0.4648, 0.4811. The corresponding final Richardson residuals are approximately  $-1.98 \times 10^{-3}$  and  $-5.54 \times 10^{-3}$ . Thus the numerical illustrations span four orders of magnitude in  $u$ . The refinement (42) also explains the slow approach to the bare logarithmic law: after subtracting its logarithmic and constant terms, the product with  $\sqrt{u}$  tends to  $-\sqrt{\pi/2}$ , while the subsequent terms are  $1/(2u)$ ,  $-\sqrt{\pi/2} u^{-3/2}/6$ , and  $1/(12u^2)$ .

All reference values here use the offset convention  $\text{Li}(x) = \text{li}(x) - \text{li}(2)$ . A software library that returns the principal-value function  $\text{li}(x) = \text{Ei}(\log x)$  must therefore have  $\text{li}(2) \approx 1.04516$  subtracted before comparison. Omitting that offset creates a false relative-error floor and can appear to violate (S21).

## S7 A renormalized zero-window identity

**Remark S7.1** (Renormalized zero-window limit and the classical Ei series). There is a separate, elementary limiting identity at the opposite end of the window parameter. For  $0 < \alpha < 1$ , define

$$\mathcal{L}_\alpha(x) := (1 - \alpha)x^\alpha \int_0^1 \frac{x^{(1-\alpha)t}}{\alpha + (1-\alpha)t} dt - (1 - \alpha)2^\alpha \int_0^1 \frac{2^{(1-\alpha)t}}{\alpha + (1-\alpha)t} dt + \log\left(\frac{\log x}{\log 2}\right). \quad (\text{S27})$$

The substitution  $s = \alpha + (1 - \alpha)t$  reduces this to

$$\mathcal{L}_\alpha(x) = \log\left(\frac{\log x}{\log 2}\right) + \int_\alpha^1 \frac{x^s - 2^s}{s} ds, \quad \mathcal{L}_\alpha(2) = 0. \quad (\text{S28})$$

The singularities of the two terms at  $s = 0$  cancel, and the exact error is

$$\text{Li}(x) - \mathcal{L}_\alpha(x) = \int_0^\alpha \frac{x^s - 2^s}{s} ds = \alpha \log\left(\frac{x}{2}\right) + O_x(\alpha^2). \quad (\text{S29})$$

For  $x \geq 2$  the error is positive and satisfies the explicit bounds

$$\alpha \log\left(\frac{x}{2}\right) \leq \text{Li}(x) - \mathcal{L}_\alpha(x) \leq \alpha x^\alpha \log\left(\frac{x}{2}\right). \quad (\text{S30})$$

Consequently  $\mathcal{L}_\alpha(x) \uparrow \text{Li}(x)$  as  $\alpha \downarrow 0$ , and

$$\text{Li}(x) = \log\left(\frac{\log x}{\log 2}\right) + \int_0^1 \frac{x^s - 2^s}{s} ds = \log\left(\frac{\log x}{\log 2}\right) + \sum_{n \geq 1} \frac{(\log x)^n - (\log 2)^n}{n n!}. \quad (\text{S31})$$

The last equality is the classical power series for  $\text{Ei}(\log x) - \text{Ei}(\log 2)$  in the offset convention used here; see [2, Eq. 6.6.1]. It is not claimed as a new series. The Ramanujan–Soldner constant is the distinguished positive zero of the principal-value logarithmic integral; Ramanujan’s accelerated series is a different expansion. The point of (S27)–(S31) is instead that the vanishing-window family approaches the classical identity after the necessary finite logarithmic counterterm.

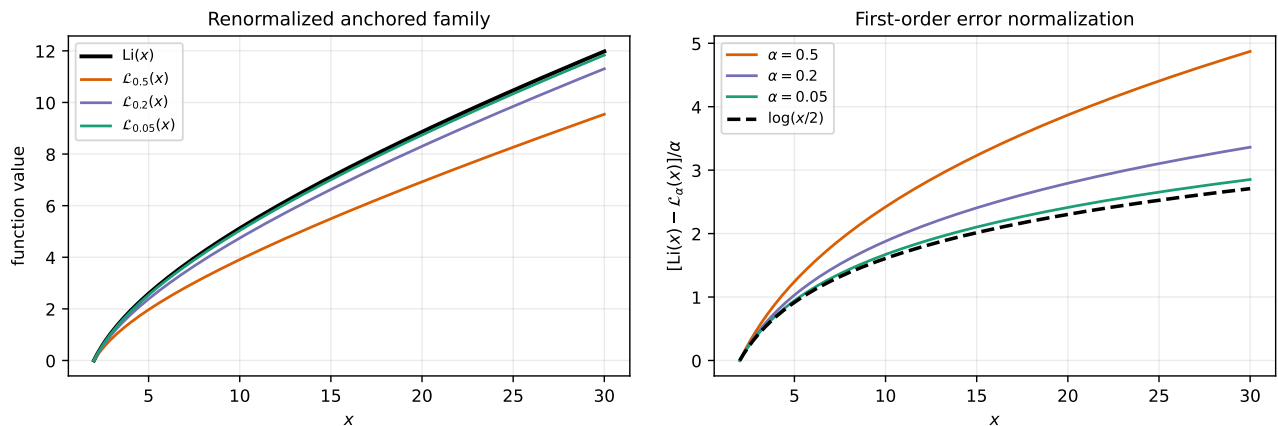


Figure S6: The renormalized anchored identity in (S27). Left:  $\mathcal{L}_\alpha(x)$  approaches the offset logarithmic integral from below as  $\alpha \downarrow 0$ . Right: the normalized error converges to  $\log(x/2)$ , confirming the first-order law in (S29). This is a diagnostic illustration of a limit leading to the classical Ei power series, not a claim of a new evaluation formula.

Nor does the diagonal limit assign a canonical sum to the divergent fixed- $a$  outer series. Such an interpretation would require a specified summability method and proof that its value agrees with the diagonal limit. No such assertion is needed here.

## S8 Numerical and interpretive considerations

The exact formulas contain factorials and alternating moments that can overflow or cancel in naive floating-point arithmetic. Stable implementations should evaluate products and moments recursively or

at increased precision. The proof of convergence is algebraic and analytic; it does not guarantee numerical stability of every syntactically equivalent implementation.

The bound (S21) is especially useful because it is relative to the positive integral itself and requires no unknown derivative constants. It is also conservative: the final clipped interval generally has a smaller ratio than  $q$ , and segmentwise bounds can be summed for a tighter certificate.

The multiscale construction is not presented as the fastest general-purpose evaluation of  $\text{Ei}(L)$ . The classical convergent series

$$\text{Ei}(L) = \gamma + \log L + \sum_{k \geq 1} \frac{L^k}{k \, k!}$$

has positive terms and is often computationally preferable; for example, it needs only on the order of  $L = \log x$  significant terms. The distinctive advantage here is instead certification: Corollary S4.2 gives alternating upper and lower enclosures together with a global relative bound, without derivative estimates or an external remainder calculation.

A classical continued-fraction comparison does not provide the same enclosure for the present positive argument. For  $z > 0$ , the function  $e^z E_1(z) = \int_0^\infty e^{-t} (z+t)^{-1} dt$  is a Stieltjes transform, so its continued fraction inherits the usual alternating-convergent bracketing. By contrast,  $\text{Ei}(L)$  for  $L > 0$  lies across the opposite Stokes direction: the corresponding Borel singularity is on the positive integration axis, and the positivity argument behind the Stieltjes enclosure does not transfer. The classical continued fraction for the generalized exponential integral is recorded in DLMF §8.19(vii) [6].

The same distinction applies to the coupled diagonal sequence. It is a structural device for turning a divergent fixed-parameter triangular family into a genuine limit and for exposing the near-threshold Borel scaling law. It is not an efficient way to compute that limit once the limiting incomplete-gamma series is known:  $T_N(x; N)$  has triangular cost and only first-order fixed- $x$  convergence, whereas the limiting series is a direct one-dimensional expansion.

Geometric scaling also prevents exact reuse of the segment moments because  $\Delta_j = (1-a)a^j U$  changes with  $j$ . Equal- $\Delta$  splitting would reuse every  $H_k(\Delta)$  but would sacrifice the uniform local ratio. A promising hybrid is therefore to use equal logarithmic steps near the top of the interval, where the lower endpoints are large, and switch to geometric steps near  $\log 2$ , where uniform contraction is most valuable. Its optimal transition point and certified work bound remain open.

## S9 Summary of the supplementary role

The results collected here complete three secondary parts of the picture. The finite-window limit admits elementary nested enclosures and an exact positive incomplete-gamma tail; geometric localization produces a separate certified multiscale expansion; and the numerical formulas make the main article's transition claims independently reproducible. The zero-window construction is retained only as a consistency limit leading to the classical offset Ei identity. None of these developments is required for the central Borel, shifted-Gaussian, or crossover theorems, which is why they are separated from the main article.

## A Auxiliary estimates

**Lemma A.1** (Endpoint moments). *Let  $b > 0$  and let  $f$  be bounded and measurable on  $[0, b]$ , continuous at  $b$ , and  $f(b) \neq 0$ . Then*

$$\int_0^b t^n f(t) dt \sim \frac{f(b)b^{n+1}}{n+1}.$$

*Proof.* After  $t = bs$ ,

$$\frac{n+1}{b^{n+1}} \int_0^b t^n f(t) dt = (n+1) \int_0^1 s^n f(bs) ds.$$

The measures  $(n+1)s^n ds$  have total mass one and concentrate at  $s = 1$ . Given  $\varepsilon > 0$ , continuity supplies  $\delta > 0$  such that  $|f(bs) - f(b)| < \varepsilon$  on  $1 - \delta < s \leq 1$ . The complementary contribution is bounded by a constant times  $(1 - \delta)^{n+1}$ , while the terminal contribution differs from  $f(b)$  by at most  $\varepsilon$ . The normalized integral therefore tends to  $f(b)$ .  $\square$

**Lemma A.2** (Finite denominator expansions). *For  $z \neq 1$  and  $z \neq -1$ , respectively,*

$$\begin{aligned} \frac{1}{1-z} &= \sum_{k=0}^N z^k + \frac{z^{N+1}}{1-z}, \\ \frac{1}{1+z} &= \sum_{k=0}^N (-1)^k z^k + (-1)^{N+1} \frac{z^{N+1}}{1+z}. \end{aligned}$$

*Proof.* Multiply the first finite sum by  $1 - z$  and the second by  $1 + z$ . The identities follow without any limiting argument. This is why every remainder formula in the paper is exact before a limit is taken.  $\square$

## B Reproducibility formulas

For independent numerical checks, the incomplete-gamma moments at integer order can be evaluated without special-function software:

$$\gamma(1, 1) = 1 - e^{-1}, \quad \gamma(m+2, 1) = (m+1)\gamma(m+1, 1) - e^{-1}. \quad (\text{S32})$$

For forward floating-point computation at large  $m$ , direct quadrature or a scaled recurrence is preferable because (S32) subtracts nearby quantities.

The positive diagonal coefficients are evaluated directly from (7); their products can be updated by

$$P_{N,0}(j) = 1, \quad P_{N,m+1}(j) = P_{N,m}(j) \frac{j+m+1}{N}.$$

For the moving-window double-scaling check, an exact  $O(N)$  probability-ratio recurrence avoids forming the triangular sum. Put

$$\mu = 1/\lambda, \quad z = (\mu + \delta)^{-1},$$

and initialize

$$Q_z(1) = 1, \quad \rho_{N,\mu}(0) = e^\mu \left(1 - \frac{\mu}{N}\right)^{N+1}.$$

Then, for  $1 \leq r \leq N+1$ ,

$$Q_z(r) = 1 + \frac{Q_z(r-1)}{z(r-1)} \quad (r \geq 2), \quad \rho_{N,\mu}(r) = \rho_{N,\mu}(r-1) \frac{N+2-r}{N-\mu},$$

and  $\rho_{N,\mu}(r) = 0$  for  $r > N+1$ . Equation (52) becomes the numerically stable exact sum

$$\frac{e^{1/\lambda}}{x} (T_N(x; \lambda N) - D_\lambda(x)) = \sum_{r \geq 1} \frac{Q_z(r)(\mu z)^r}{r} (\rho_{N,\mu}(r) - 1). \quad (\text{S33})$$

The geometrically decaying tail beyond  $N+1$  is continued until the requested precision is reached. This recurrence produced the moving-window residual ratios reported in Remark 5.17.

At extended precision the multiplicative recurrence above is stable for the tests in this paper. In ordinary double precision, however, the small- $r$  difference  $\rho_{N,\mu}(r) - 1 = O(r^2/N)$  can lose roughly  $\log_{10} N$  digits. A cancellation-resistant implementation accumulates the logarithm instead:

$$\ell_0 = \mu + (N + 1) \log 1p(-\mu/N), \quad \ell_r = \ell_{r-1} + \log(N + 2 - r) - \log(N - \mu),$$

for  $1 \leq r \leq N + 1$ , and evaluates the difference in (S33) as  $\text{expm1}(\ell_r)$ . The printed multiplicative recurrence also self-terminates: its next numerator is zero at  $r = N + 2$ , after which  $\rho_{N,\mu} = 0$  and only the geometrically decaying comparison tail remains.

For the double-scaling targets in Table S3, the exact derivative identity in Proposition 5.14 gives the bounded-integrand representation

$$\mathcal{B}\Phi(u) = \int_0^u \left[ -\frac{1}{2} + \frac{1}{2} \sqrt{\frac{\pi}{2v}} e^{1/(2v)} \text{erfc}\left(\frac{1}{\sqrt{2v}}\right) \right] \frac{dv}{v}.$$

The bracket is  $O(v)$  at the origin, so the displayed integrand remains bounded. As an independent cross-check, the equivalent positive-axis formula

$$\mathcal{B}\Phi(u) = \int_0^\infty (e^{-u\rho^2/2} - 1) e^{-\rho} \frac{d\rho}{\rho}$$

has an integrand that is  $O(\rho)$  at the origin and is exponentially damped at infinity. The bounded limit in the erfc formula is not obtained reliably by multiplying  $e^{1/(2v)}$  and  $\text{erfc}(1/\sqrt{2v})$  as separately evaluated floating-point numbers: near  $v = 0$  the first overflows while the second underflows, and adaptive quadrature can silently replace the bounded integrand by a spurious  $-1/(2v)$  singularity. Stable code must evaluate the scaled complementary error function  $\text{erfcx}(z) = e^{z^2} \text{erfc}(z)$  directly, or use its standard large- $z$  asymptotic expansion. For  $u \gtrsim 1$ , the exact convergent expansion (42) is generally the simplest high-precision evaluation: its terms require only gamma values and elementary powers. For small  $u$ , those alternating terms can grow substantially before decaying. With the scaled evaluation just described, both integral formulas are stable alternatives, but in numerical tests the exponentially damped positive-axis integral often reaches a requested small- $u$  accuracy with fewer quadrature refinements than the erfc form. The multiscale moments

$$H_k(\Delta) := \int_0^\Delta t^k e^t dt$$

satisfy

$$H_0(\Delta) = e^\Delta - 1, \quad H_k(\Delta) = \Delta^k e^\Delta - k H_{k-1}(\Delta). \quad (\text{S34})$$

As with (S32), cancellation can occur when  $k$  is large relative to  $\Delta$ ; compensated arithmetic, backward recurrences, or quadrature may then be appropriate.

The numerical figure in this paper used the definitions exactly as printed, with high-precision quadrature for reference values. In particular, software returning the principal-value  $\text{li}(x) = \text{Ei}(\log x)$  must be converted to this paper's offset convention by subtracting  $\text{li}(2) = \text{Ei}(\log 2) \approx 1.04516$ . Otherwise the multiscale comparison acquires a spurious error floor of  $\text{li}(2)/\text{Li}(x)$ ; at  $x = 10^6$  this is approximately  $1.33 \times 10^{-5}$  and can falsely appear to violate (S21).

The formulas above, rather than the graphing notebooks, are the reproducibility specification for the numerical values in the paper. The four exploratory graphing-calculator notebooks that motivated the investigation are available at:

- lower incomplete-gamma expansion;
- two expansions for  $\text{Li}(x) - \text{Li}(x/e)$ ;
- triangular finite transformations;
- recursive and multiscale expansions.

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